

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

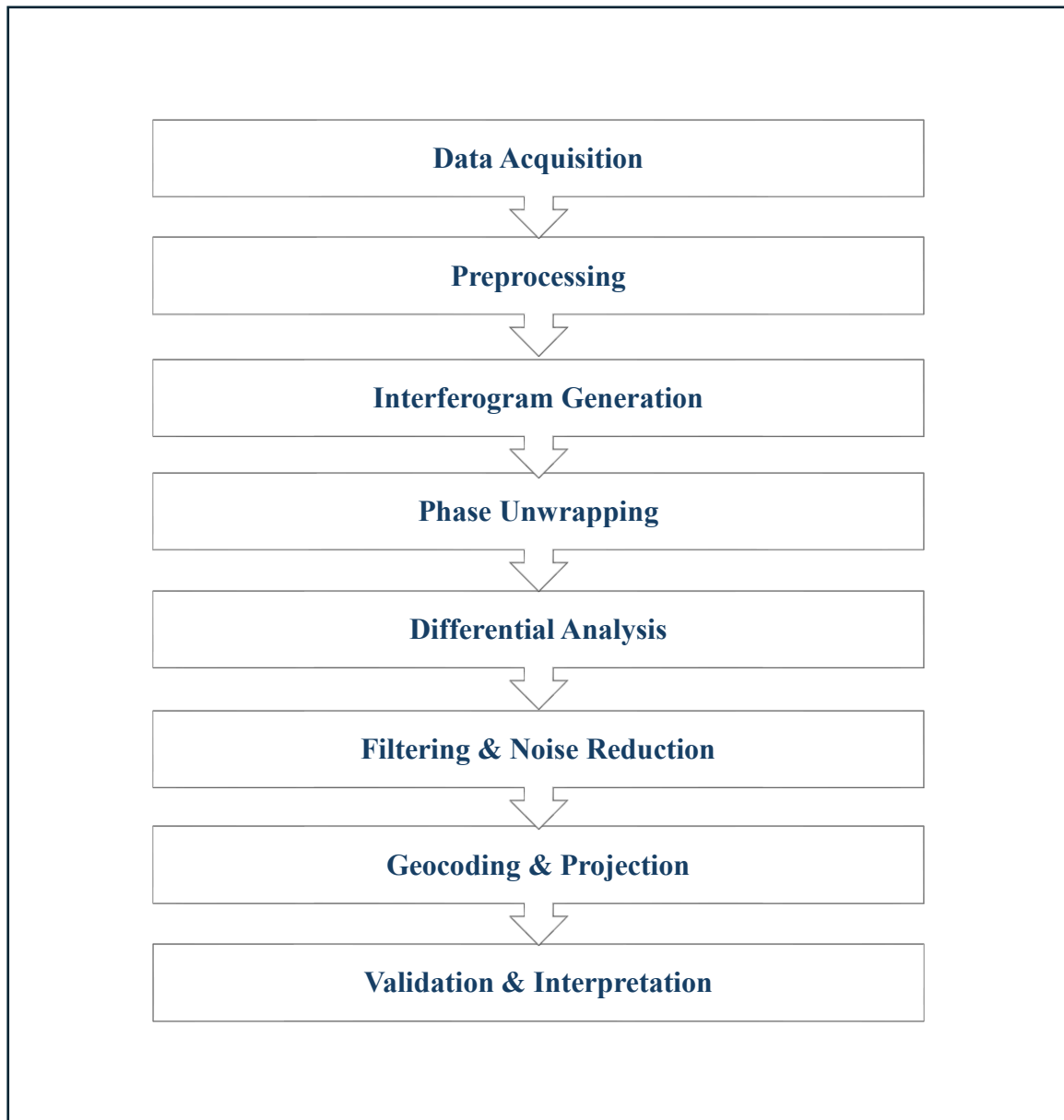
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 1002+580

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

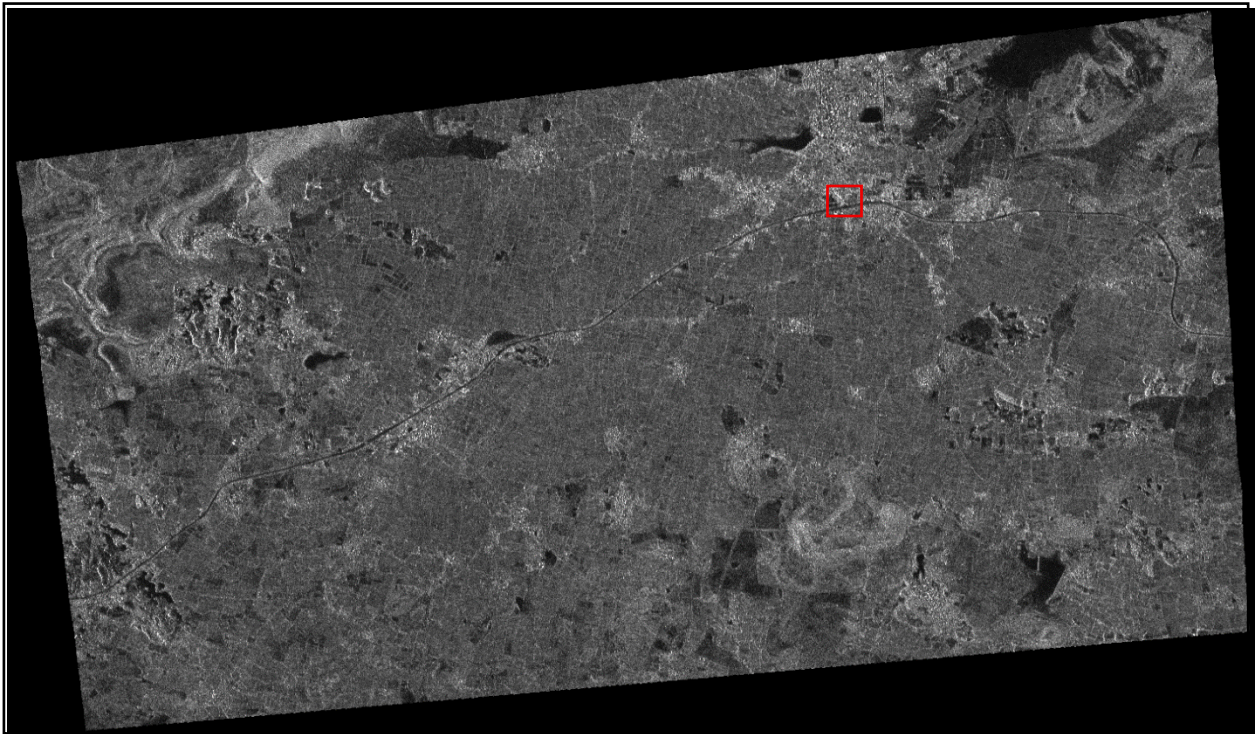
Flight Direction - DESCENDING

Polarization – VV+VH

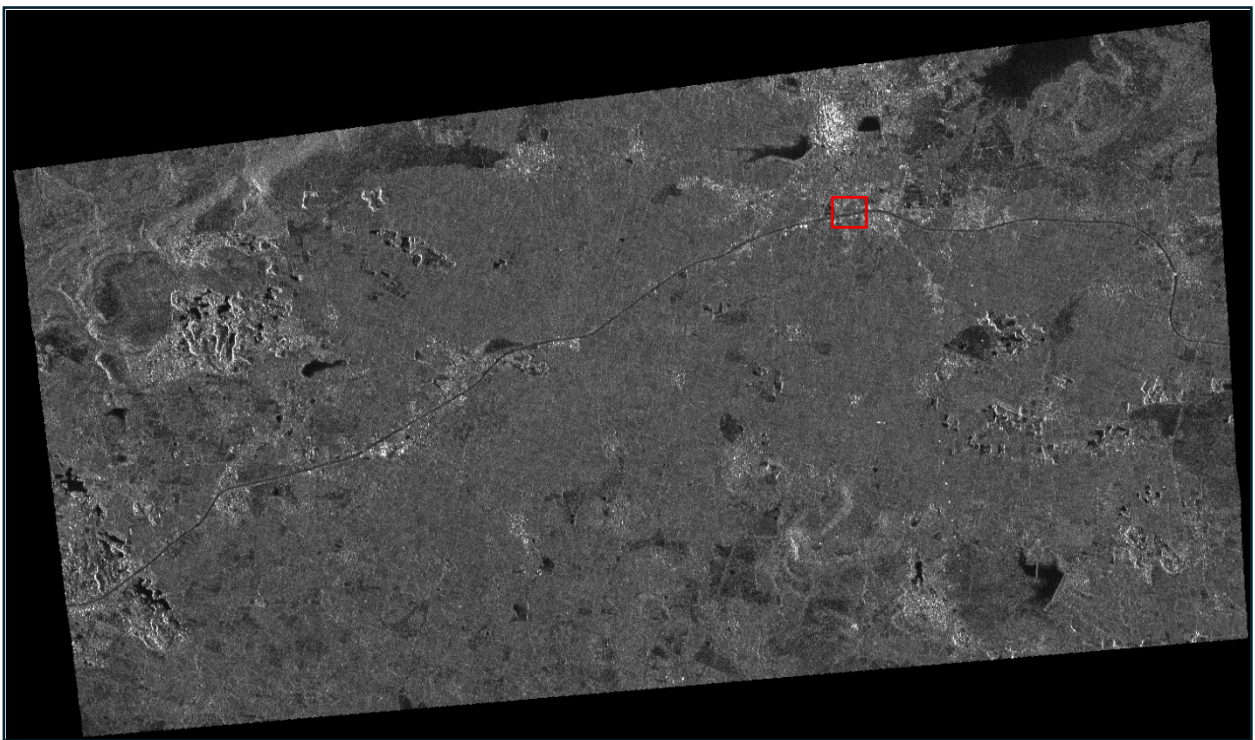
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

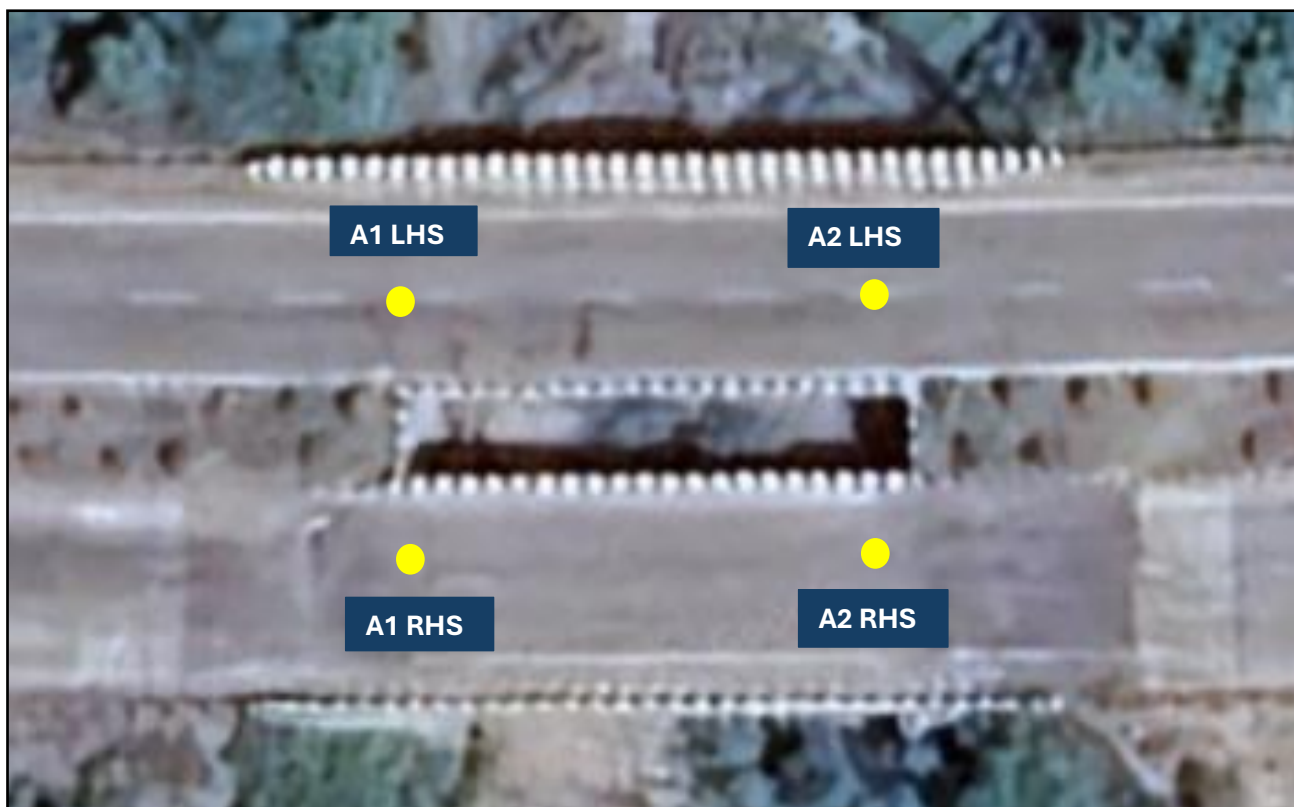


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

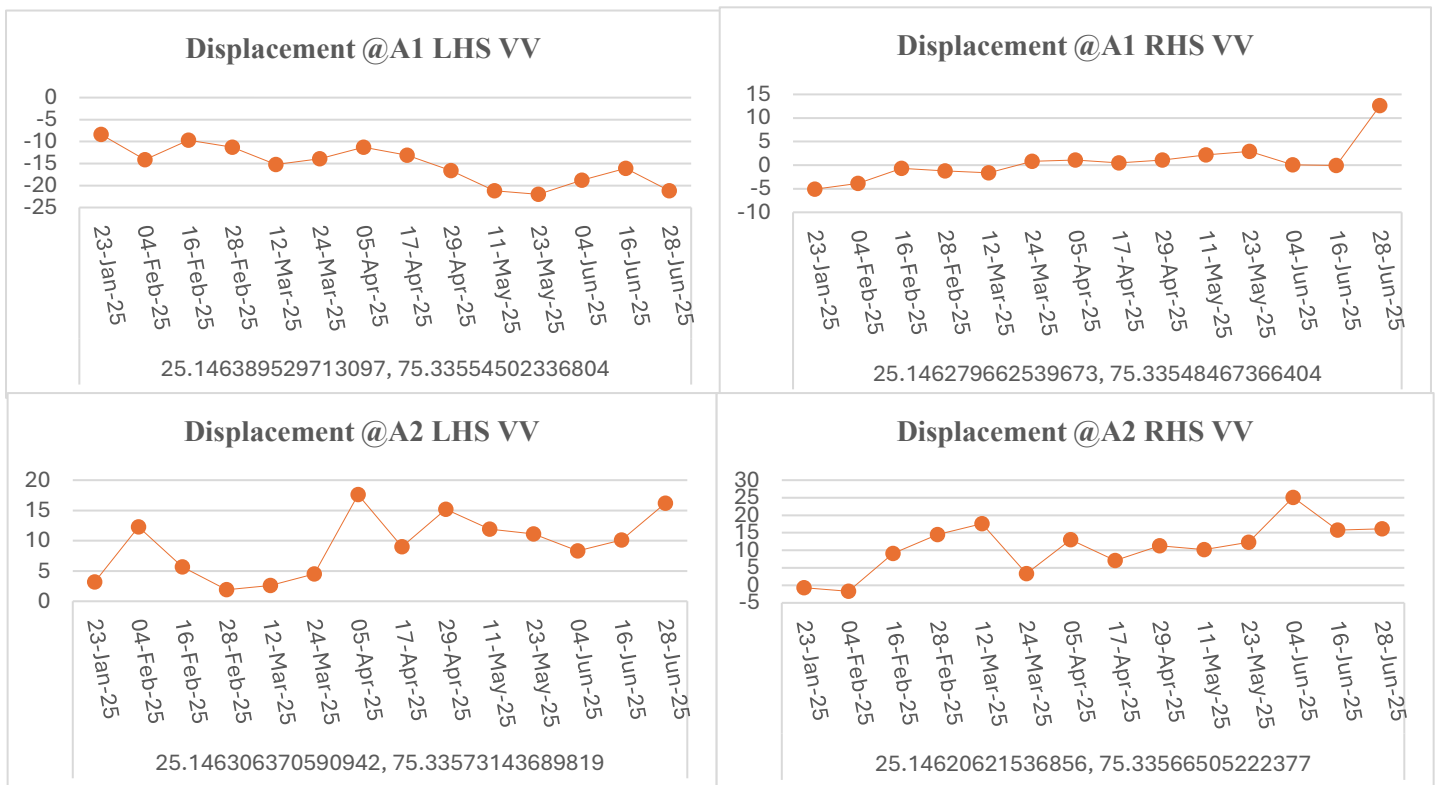
MNB – CH 1002+580

Length – 34.66M

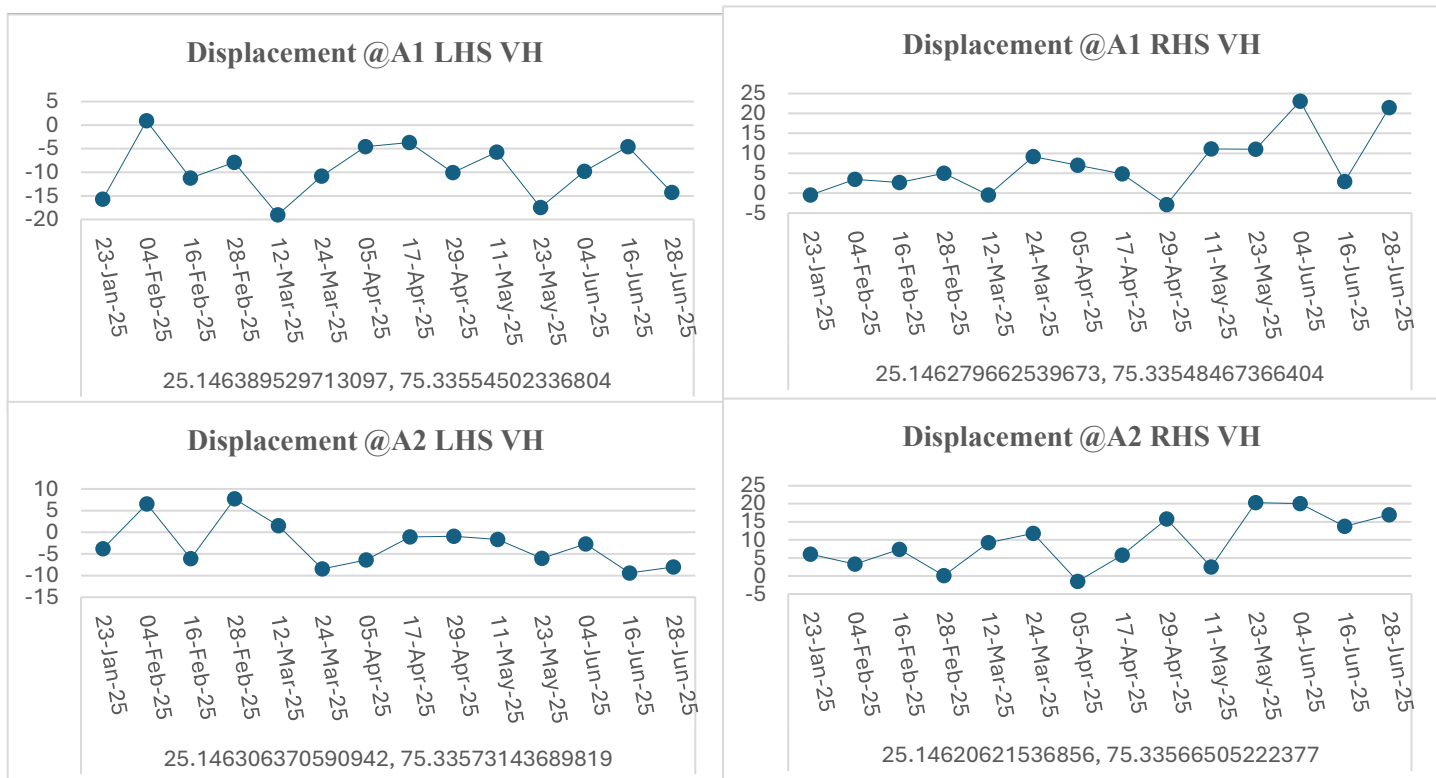


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



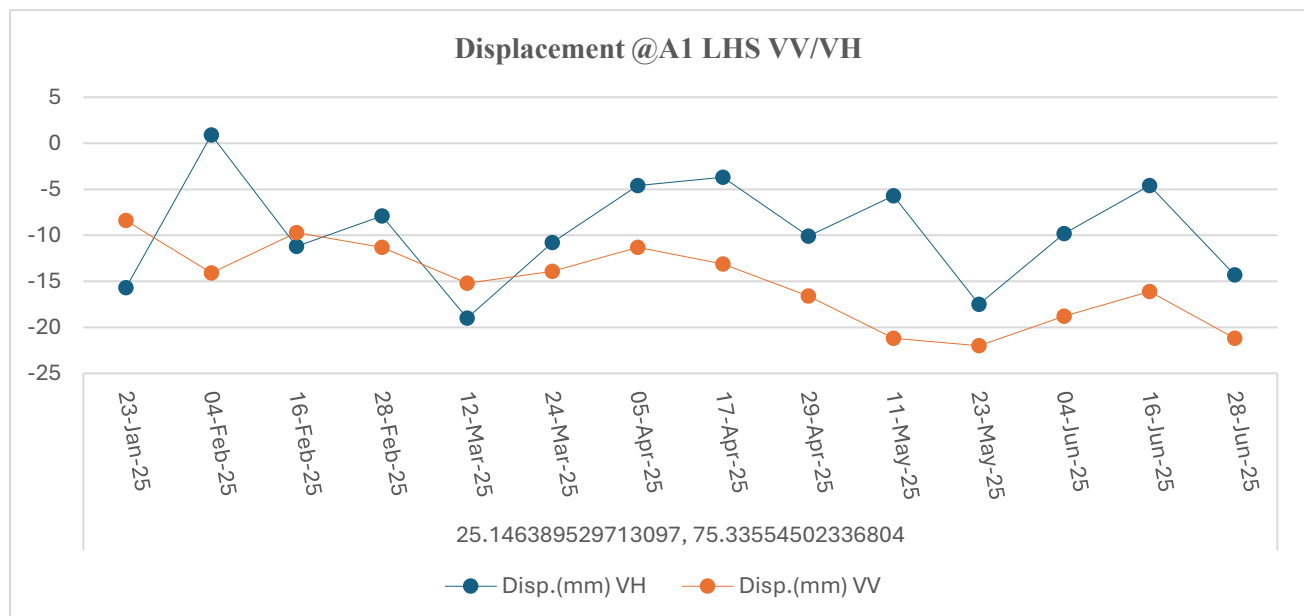
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.146389529713097, 75.33554502336804)



Key Observations:

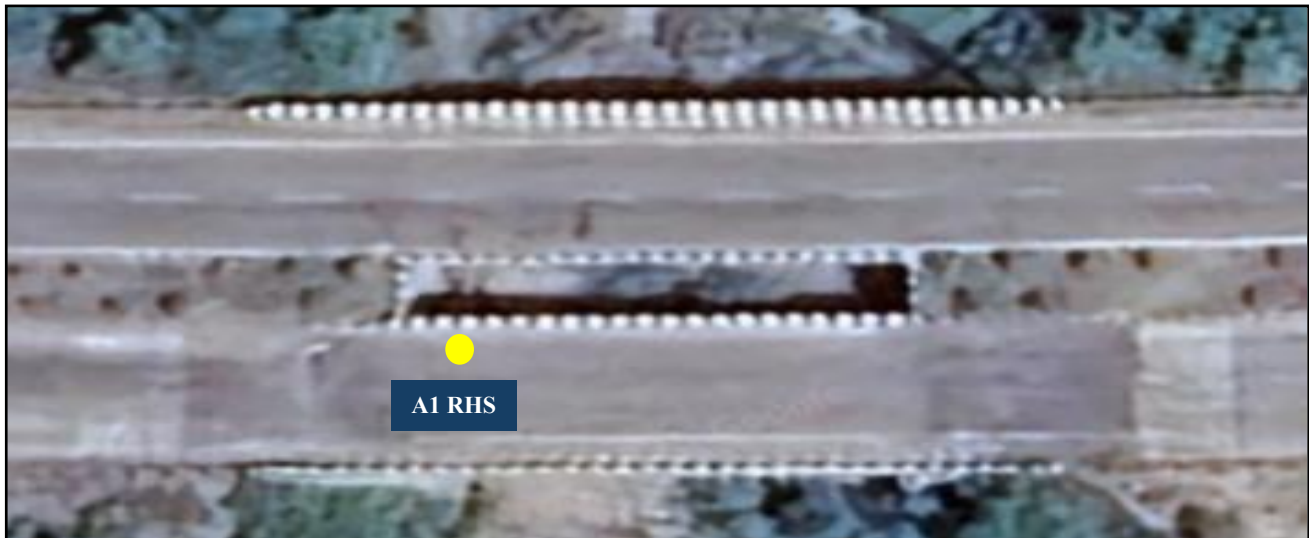
Displacement Magnitude Patterns:

- Vertical displacement (VV)** Vertical displacement remains consistently negative throughout the monitoring period (Jan–Jun 2025), indicating persistent settlement.
- Horizontal Displacement (VH)** Horizontal displacement exhibits high variability with strong oscillatory behaviour, ranging approximately between –18 mm and +2 mm.

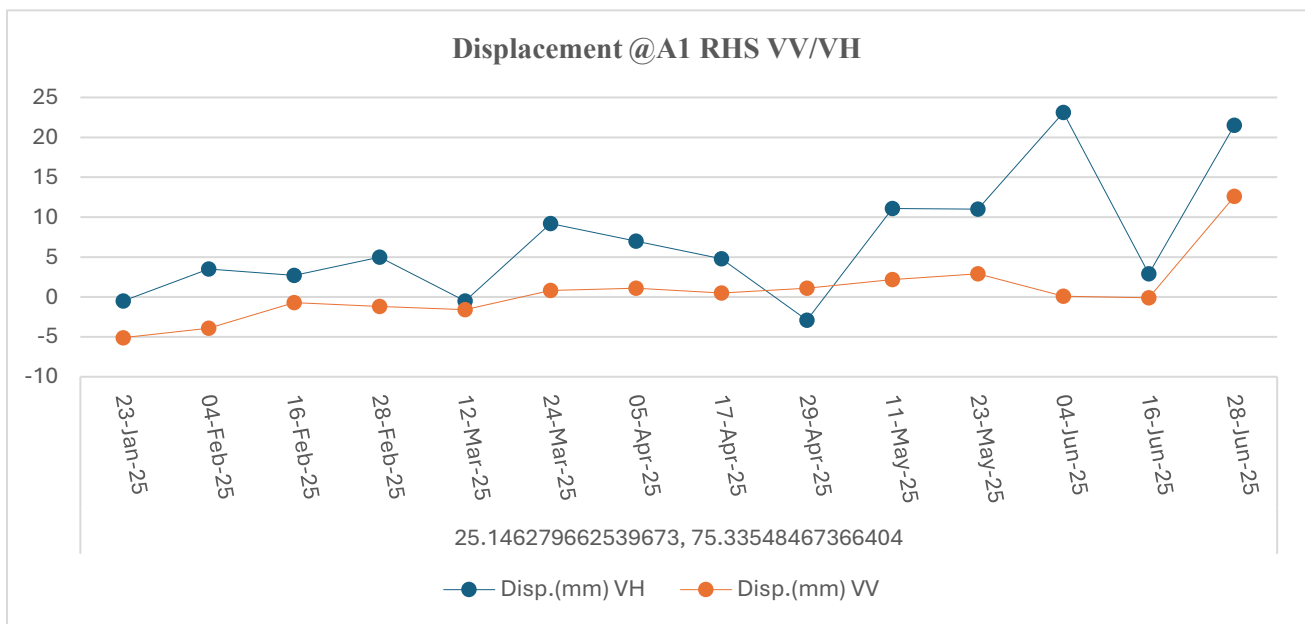
Critical Events & Anomalies:

- February 2025:** VH shows a brief positive peak (+2 mm) while VV remains negative (–14 mm), indicating a short-lived lateral uplift event without vertical recovery.
- May 2025:** VV reaches its maximum settlement (–22 mm), while VH also drops sharply (–17 mm), representing the most critical deformation phase in the monitoring period
- June 2025:** VH shows a partial rebound toward –5 mm, while VV improves slightly but remains negative (–16 to –18 mm), indicating post-event adjustment rather than full recovery.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.146279662539673, 75.33548467366404)



Key Observations:

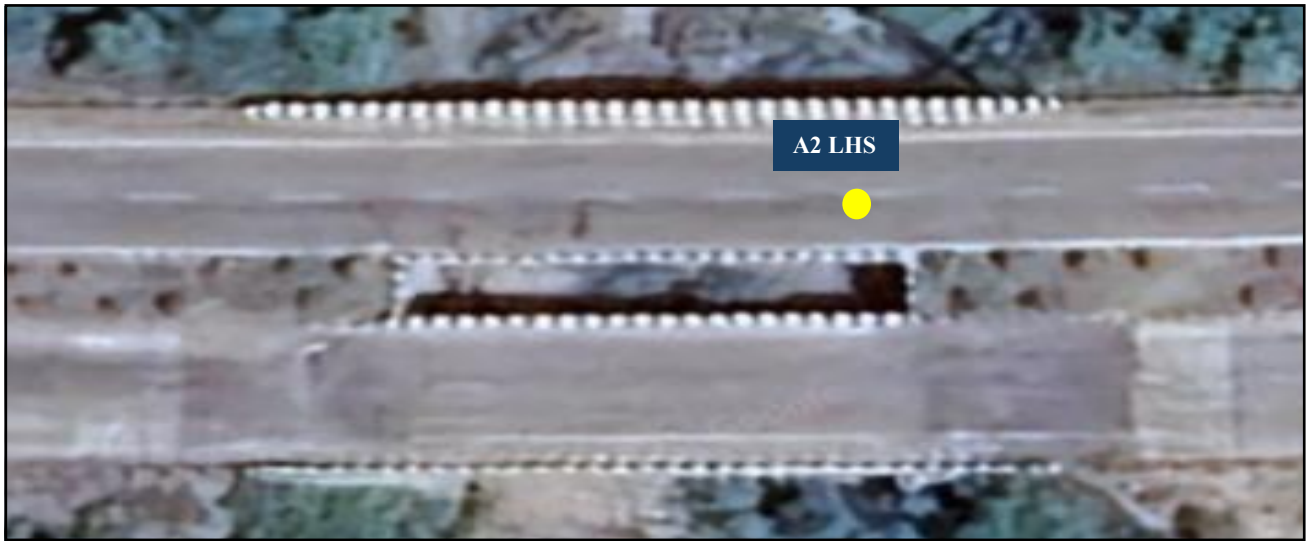
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits high variability with strong positive excursions, ranging roughly from –3 mm to +23 mm.
- **Vertical displacement (VV)** Vertical displacement shows low to moderate variability, fluctuating between approximately –5 mm and +12 mm over the monitoring period (Jan–Jun 2025).

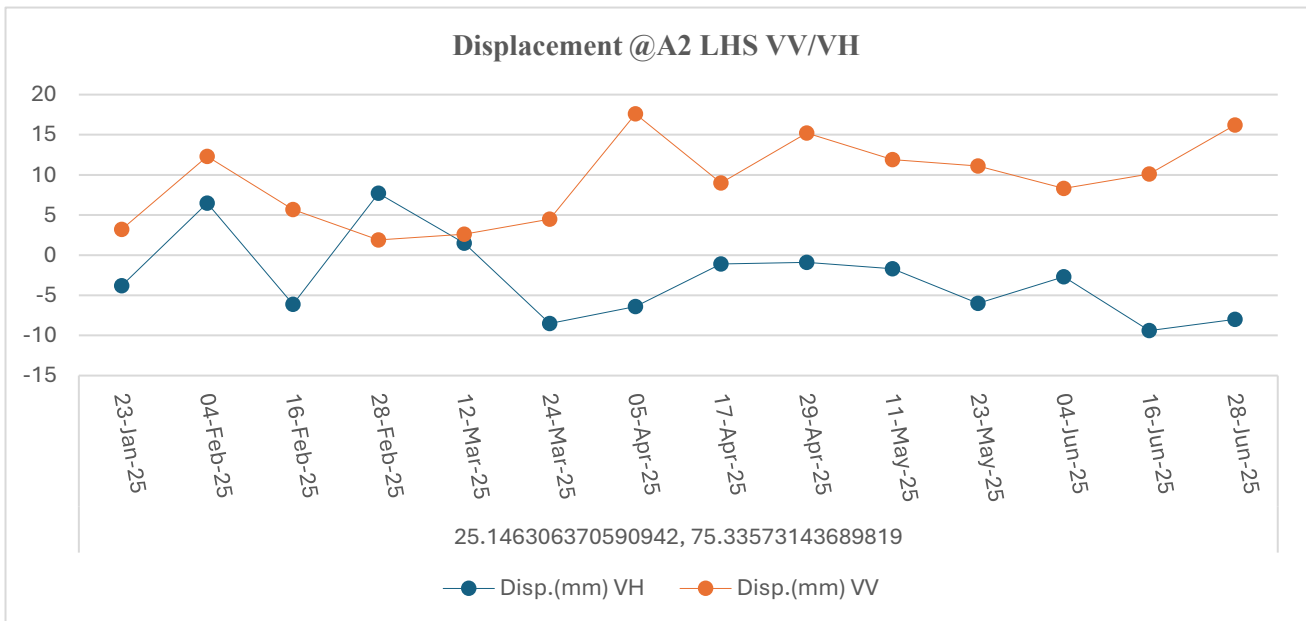
Critical Events & Anomalies:

- **January–February 2025:** VH increases from near 0 mm to +4 mm, while VV remains negative (–5 to –3 mm), indicating early lateral movement without vertical uplift, possibly related to surface or structural adjustment
- **June 2025:** Both VV and VH increase sharply (VV +12 mm, VH +21 mm), indicating simultaneous vertical uplift and lateral displacement intensification, representing the most significant combined deformation event during the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.146306370590942, 75.33573143689819)



Key Observations:

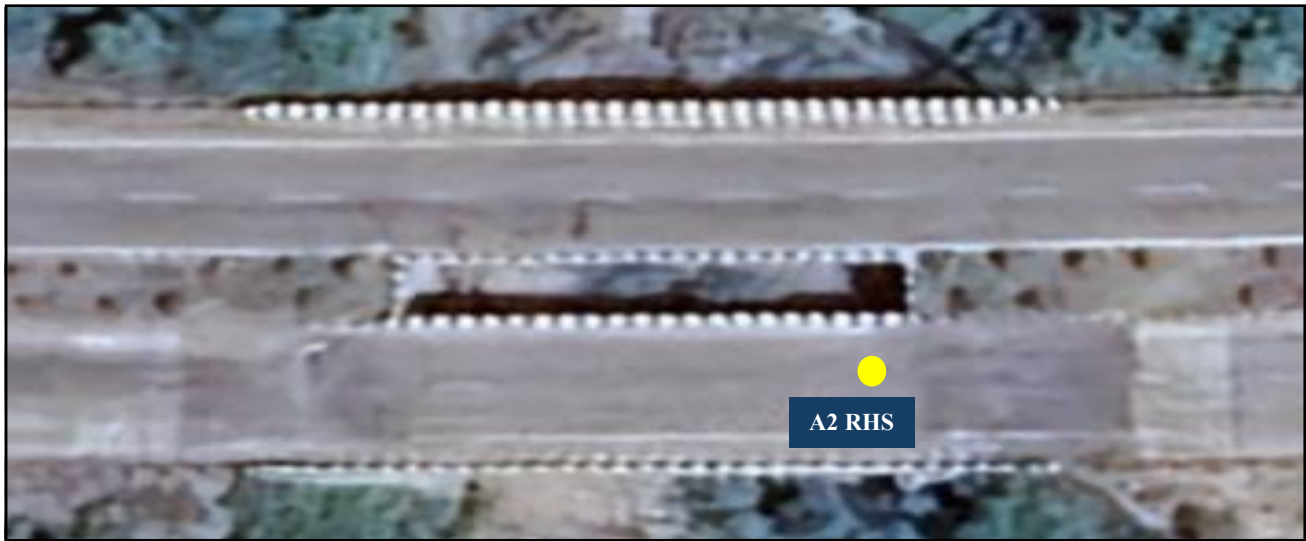
Displacement Magnitude Patterns:

- **Vertical displacement (VV)** Vertical displacement remains predominantly positive throughout the monitoring period (Jan–Jun 2025), indicating sustained vertical uplift.
- **Horizontal Displacement (VH)** Horizontal displacement is consistently negative, ranging roughly between –10 mm and +8 mm, but dominated by negative values after mid-March.

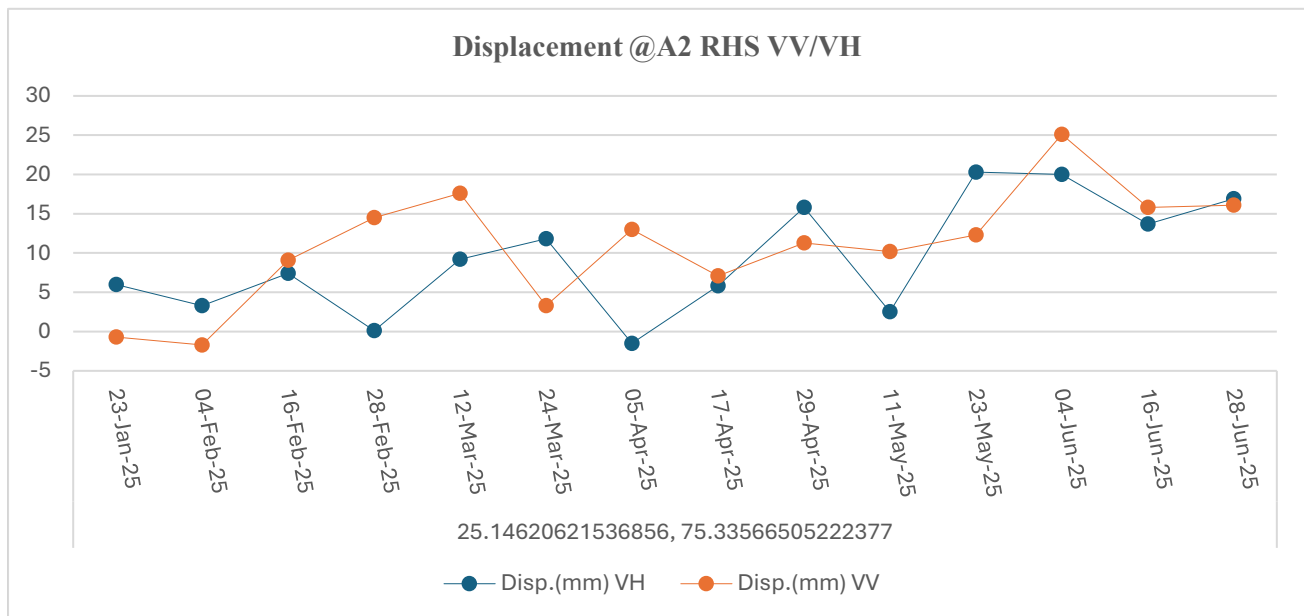
Critical Events & Anomalies:

- **February 2025:** VH reaches a local positive peak (+6 mm) while VV rises sharply to +12 mm, indicating a short-term coupled vertical uplift and lateral expansion event.
- **June 2025:** VV rises once more to +16 mm, while VH stays strongly negative (–8 to –9 mm), marking a persistent uplift-dominated regime with continued lateral displacement.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.14620621536856, 75.33566505222377)



Key Observations:

Displacement Magnitude Patterns:

- Vertical (VV) displacement** Vertical displacement shows a clear long-term upward trend, transitioning from slight negative/near-zero values in January to strong positive uplift by June 2025.
- Horizontal (VH) displacement** Horizontal displacement exhibits moderate to high variability, fluctuating between roughly -2 mm and +20 mm.

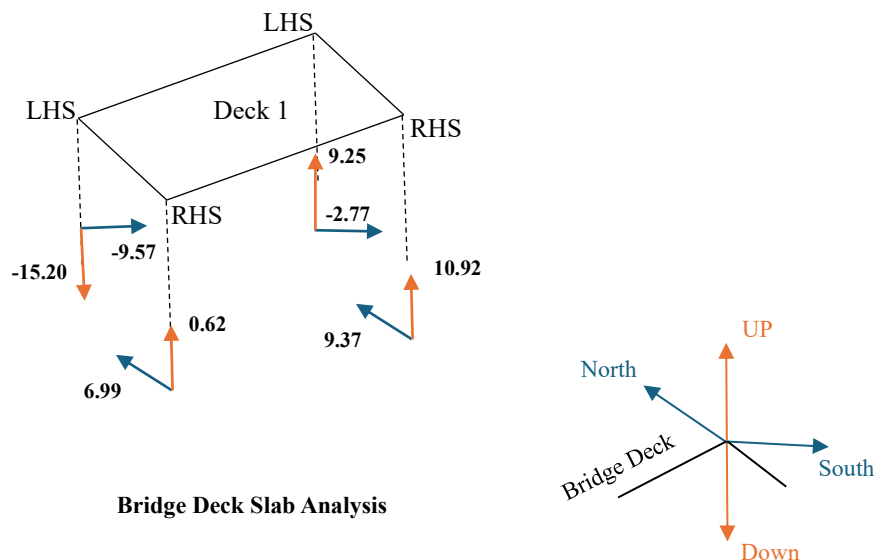
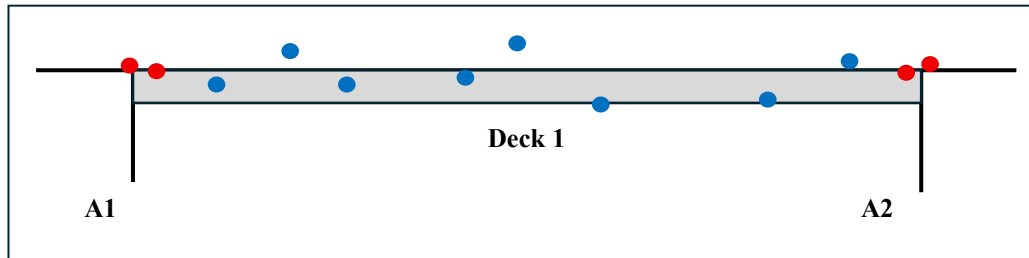
Critical Events & Anomalies:

- January-February 2025:** VH decreases slightly from +6 mm to +3 mm, while VV remains near zero to slightly negative (0 to -2 mm), indicating initial lateral adjustment without vertical uplift.
- June 2025:** Both VH and VV stabilize at elevated values (VH +14-17 mm, VV +16-17 mm), suggesting post-peak adjustment under sustained deformation.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- Vertical displacements across Deck 1 range approximately between -15.20 mm (downward displacement) and $+10.92$ mm (uplift), indicating pronounced differential vertical movement across the deck. The left-hand side (LHS) shows dominant downward movement, with vertical displacement of approximately -15.20 mm, indicating localized settlement at the LHS support. The right-hand side (RHS) exhibits predominant uplift, with vertical displacement reaching approximately $+9.25$ mm to $+10.92$ mm, suggesting upward response relative to the LHS.

Horizontal Displacement Behaviour:

- Horizontal displacements range approximately between 2.77 mm and 9.57 mm, with movement directions predominantly toward the north and northwest, as indicated by the displacement vectors. Overall, lateral displacements are moderate and spatially variable, indicating non-uniform horizontal response across the deck width rather than rigid-body translation.

CONCLUSION:

- The displacement behaviour of Deck 1 indicates significant vertical differential movement, with settlement at the LHS and uplift at the RHS, suggesting rotational response of the deck slab. Overall, the deck exhibits stable but asymmetrical behaviour, and trend-based assessment is recommended rather than immediate intervention